How to Choose a Chart

A Statistically Motivated Guide

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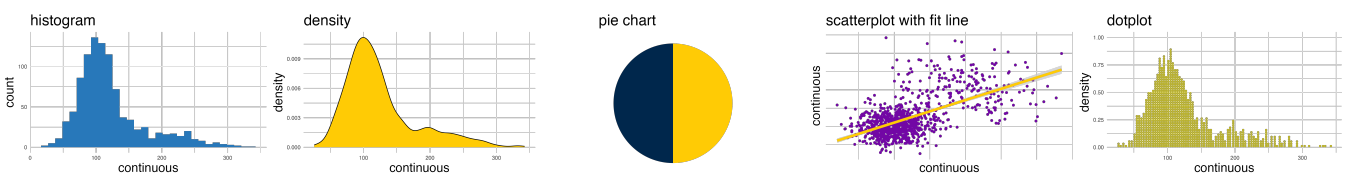
```
library(ggplot2) # beautiful graphs

library(ggthemes) # nice themes for ggplot2

library(ggbeeswarm) # helpful 'beeswarm' geometry

library(ggdist) # 'distribution' geometries

library(cowplot) # arrange graphs

library(pander) # nice tables
```



How to Choose a Chart

Choosing the right chart to represent your data can be a daunting process. I believe that a *starting point* for this thinking is some basic statistical thinking about the *type* of variables that you have. At the broadest level, variables may be conceptualized as *categorical* variables, or *continuous* variables.

- *categorical variables* represent unordered categories like *gender*, or *religious affiliation*.
- *continuous variables* represent a continuous scale like a *mental health scale*, or a *measure of neighborhood quality*.

Once we have discerned the type of variable that have, there are two followup questions we may ask before deciding upon a chart strategy:

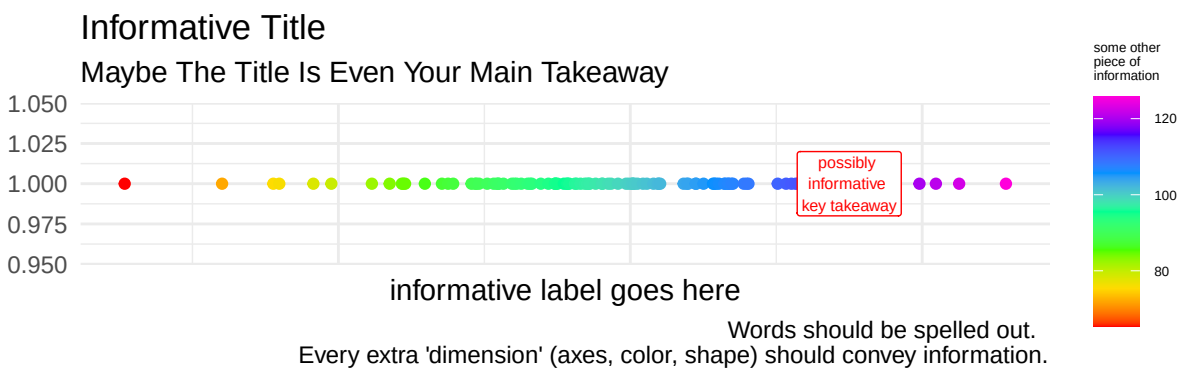
- Is our graph about **one thing at a time**?
  - How much of *x* is there?

  - What is the distribution of *x*?
- Is our graph about **two things at a time**?
  - What is the relationship of *x* and *y*?
  - How are *x* and *y* associated?

A Few Notes

A Note About Graph Labels

Graphs should have clear titles and labels.



A Note About Software

The principles of graphing discussed in this document transcend any particular software package, and could be implemented in many different software packages, such as SPSS, SAS, Stata, or R.

The graphs in these particular examples use ggplot2, a graphing library in R. `ggplot2` graph syntax can be formidably complex, with a steep learning curve. More information about ggplot can be found here.

A Note About The Code In This Document

Note that ggplot2 can be MUCH simpler than these examples make it look.

For example,

```
ggplot(mydata, aes(x = x)) + geom_histogram()
```

will produce a perfectly serviceable histogram.

Much of the complication of the code in this document is simply the result of formatting tweaks to get the graphs *EXACTLY* the way I wanted them.

Observe also, that for layout purposes, I am reading each ggplot call into an object, e.g.

```
p1 <- ggplot(mydata, aes(x = x)) + geom_histogram()
```

so that I can later use `plot_grid` to lay out the graphs.

In your own work, you do not need to do this, and it may be simpler to simply say:

```
ggplot(...) + ...
```

A Note About Graph Colors

This document uses colors based upon official University of Michigan colors. Using colors that match the design scheme of your organization may be helpful.

```
# michigan colors

michigan_colors=c("#00274c", # blue
                  "#ffcb05", # maize
                  "#a4270b", # tappan red
                  "#e96300", # ross school orange
                  "#beb300", # wave field green
                  "#21c1bc", # taubman teal
                  "#2878ba", # arboretum blue
                  "#7207a5") # ann arbor amethyst

# name individual colors

michigan_blue <- "#00274c"

michigan_maize <- "#ffcb05"

tappan_red <- "#a4270b"

ross_school_orange <- "#e96300"

wave_field_green <- "#beb300"

taubman_teal <- "#21c1bc"

arboretum_blue <- "#2878ba"

ann_arbor_amethyst <- "#7207a5"
```

A Simulated Data File of Continuous and Categorical Data

A few randomly selected observations…

	x	y	z	u	v	w	s	q
502	100.9	108.6	80.06	Group A	Group A	Group A	Group 2	120.9
652	121.8	92.37	128.1	Group A	Group A	Group A	Group 4	161.8
252	93.6	59.93	121.3	Group B	Group B	Group A	Group 2	113.6
196	67.22	99.17	67.7	Group B	Group B	Group A	Group 1	77.22
993	94.36	136.9	75.21	Group B	Group B	Group A	Group 3	124.4
140	319.9	197.7	62.06	Group B	Group B	Group B	Group 2	339.9
118	231.1	188.4	94.42	Group B	Group B	Group B	Group 2	251.1
144	98.25	108	123.1	Group B	Group B	Group A	Group 4	138.2
336	111.4	127.1	96.81	Group A	Group B	Group A	Group 3	141.4
278	104.7	73.18	158.9	Group B	Group B	Group A	Group 2	124.7

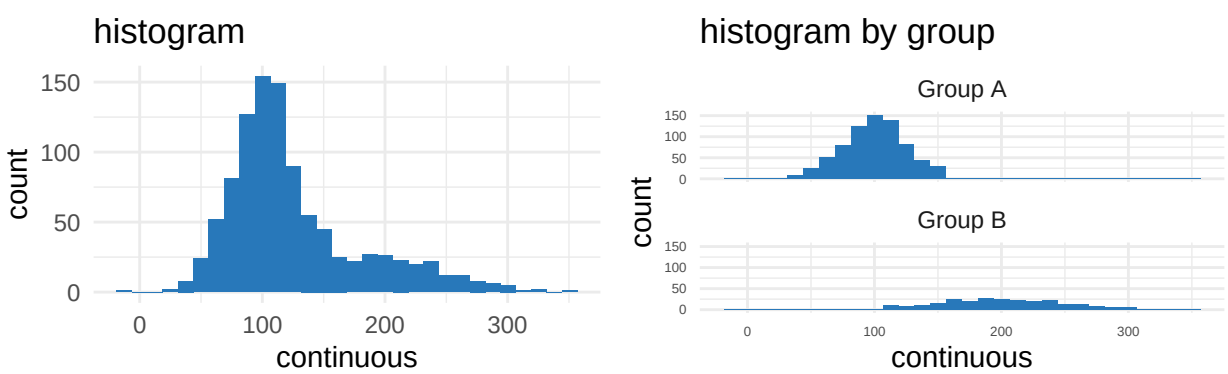
One Thing At A Time

Two Things At A Time

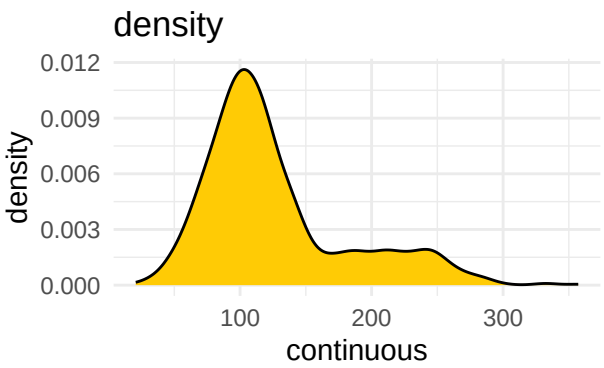
Continuous

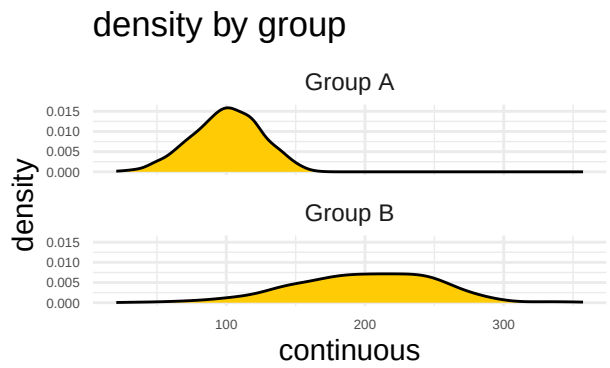
Continuous By Categorical

```
my_histogram <- ggplot(mydata, aes(x = x)) +
  geom_histogram(fill = arboretum_blue) +
  ggtitle("histogram") +
  xlab("continuous") + ylab("count") +
  theme_minimal()

my_facet_histogram <- ggplot(mydata, aes(x = x)) +
  geom_histogram(fill = arboretum_blue) +
  facet_wrap(~w, nrow = 2) +
  ggtitle("histogram by group") +
  xlab("continuous") + ylab("count") +
  theme_minimal() +
  theme(axis.text = element_text(size = 5)) # small font size for axis

plot_grid(my_histogram, my_facet_histogram, ncol=2)
```

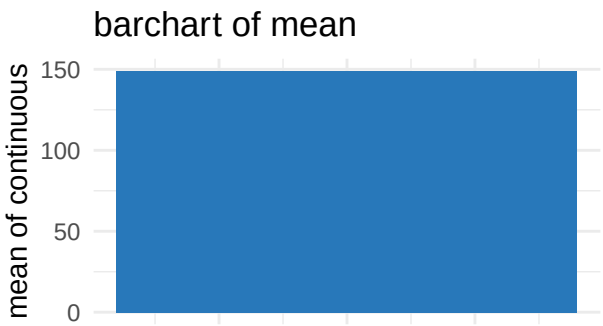


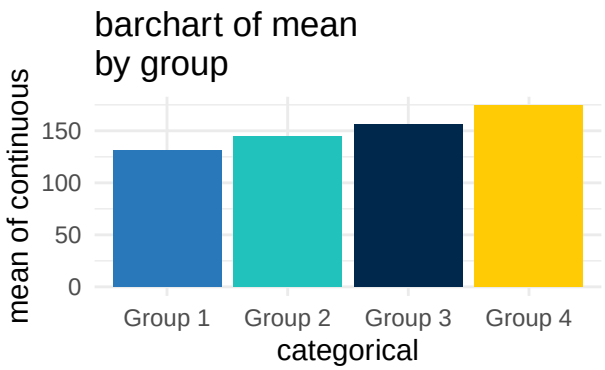

```
my_density <- ggplot(mydata, aes(x = y)) +
  geom_density(fill = michigan_maize) +
  ggtitle("density") +
  xlab("continuous") + ylab("density") +
  theme_minimal()

my_facet_density <- ggplot(mydata, aes(x = y)) +
  geom_density(fill = michigan_maize) +
  facet_wrap(~w, nrow = 2) +
  ggtitle("density by group") +
  xlab("continuous") + ylab("density") +
  theme_minimal() +
  theme(axis.text = element_text(size = 5)) # small font size for axis

plot_grid(my_density, my_facet_density, ncol = 2)
```



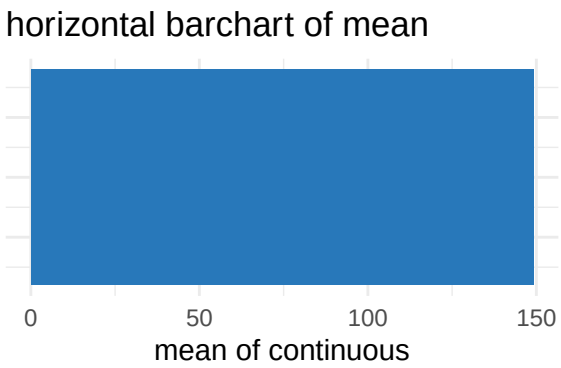


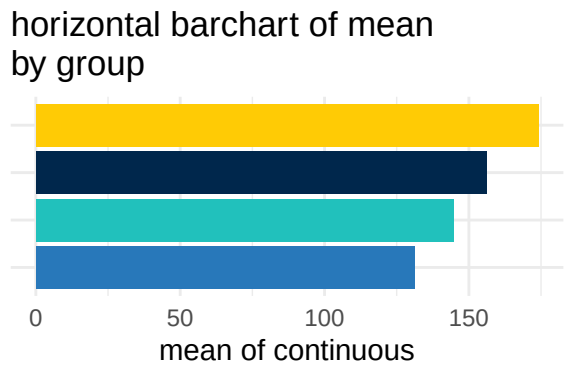

```
my_m_barchart <- ggplot(mydata,
                        aes(x = 1,
                            y = q,
                            fill = factor(1))) +
  stat_summary(fun = mean, geom = "bar") +
  scale_fill_manual(values = c(arboretum_blue)) +
  ggtitle("barchart of mean") +
  guides(fill=FALSE) +
  xlab(" ") +
  ylab("mean of continuous") +
  theme_minimal() +
  theme(axis.text.x = element_blank()) +
  theme(axis.ticks.x = element_blank())

my_facet_m_barchart <- ggplot(mydata,
                              aes(x = factor(s),
                                  y = q,
                                  fill = s)) +
  stat_summary(fun = mean, geom = "bar") +
  scale_fill_manual(values = c(arboretum_blue,
                               taubman_teal,
                               michigan_blue,
                               michigan_maize)) +
  ggtitle("barchart of mean \nby group") +
  guides(fill=FALSE) +
  xlab("categorical") + ylab("mean of continuous") +
  theme_minimal()

plot_grid(my_m_barchart,
          my_facet_m_barchart,
          ncol = 2)
```



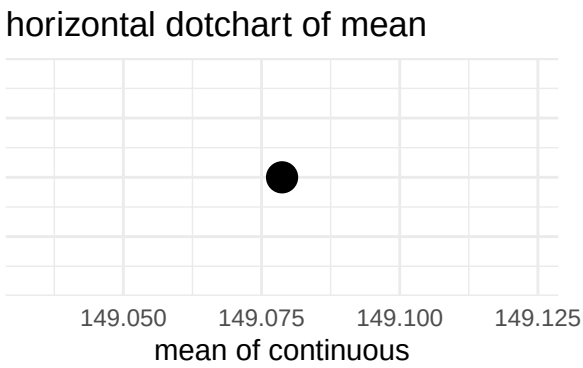


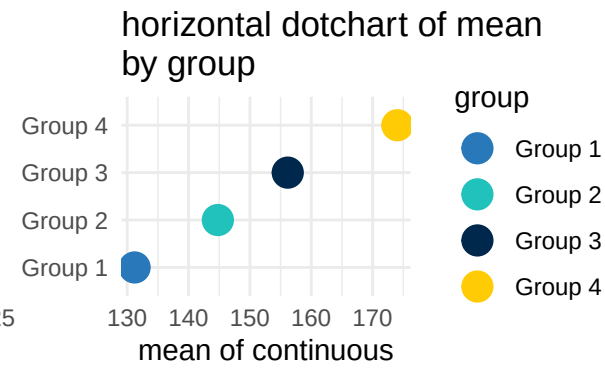

```
my_horiz_m_barchart <- ggplot(mydata,
                              aes(x = 1,
                                  y = q,
                                  fill = factor(1))) +
  stat_summary(fun = mean,
               geom = "bar") +
  coord_flip() +
  scale_fill_manual(values = c(arboretum_blue)) +
  ggtitle("horizontal barchart of mean") +
  guides(fill=FALSE) +
  xlab(" ") +
  ylab("mean of continuous") +
  theme_minimal() +
  theme(axis.text.y = element_blank()) +
  theme(axis.ticks.y = element_blank())

my_facet_horiz_m_barchart <- ggplot(mydata,
                                    aes(x = factor(s),
                                        y = q,
                                        fill = s)) +
  stat_summary(fun = mean,
               geom = "bar") +
  coord_flip() +
  scale_fill_manual(values = c(arboretum_blue,
                               taubman_teal,
                               michigan_blue,
                               michigan_maize)) +
  ggtitle("horizontal barchart of mean \nby group") +
  guides(fill=FALSE) +
  xlab(" ") + ylab("mean of continuous") +
  theme_minimal() +
  theme(axis.text.y = element_blank()) +
  theme(axis.ticks.y = element_blank())

plot_grid(my_horiz_m_barchart,
          my_facet_horiz_m_barchart)
```





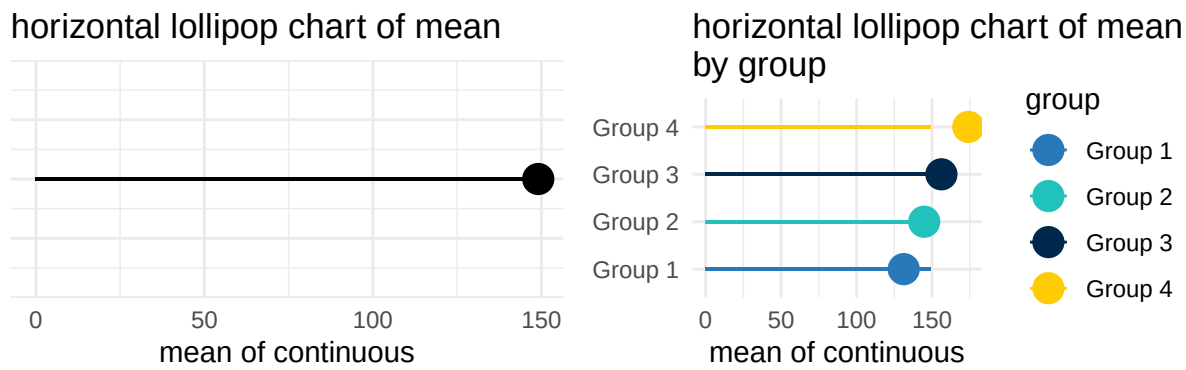

```
my_horiz_m_dotchart <- ggplot(mydata,
                              aes(x = 1,
                                  y = q,
                                  fill = factor(1))) +
  stat_summary(fun = mean,
               geom = "point", size = 5) +
  coord_flip() +
  scale_color_manual(values = c(arboretum_blue)) +
  ggtitle("horizontal dotchart of mean") +
  guides(fill = FALSE) +
  xlab(" ") +
  ylab("mean of continuous") +
  theme_minimal() +
  theme(axis.text.y = element_blank(),
        axis.ticks.y = element_blank())

my_facet_horiz_m_dotchart <- ggplot(mydata,
                                    aes(x = factor(s),
                                        y = q,
                                        color = s)) +
  stat_summary(fun = mean,
               geom = "point",
               size = 5) +
  coord_flip() +
  scale_color_manual(name = "group",
                     values = c(arboretum_blue,
                                taubman_teal,
                                michigan_blue,
                                michigan_maize)) +
  ggtitle("horizontal dotchart of mean \nby group") +
  guides(fill=FALSE) +
  xlab(" ") +
  ylab("mean of continuous") +
  theme_minimal() +
  theme(axis.title.y = element_blank(),
        axis.ticks = element_blank())

plot_grid(my_horiz_m_dotchart,
          my_facet_horiz_m_dotchart)
```



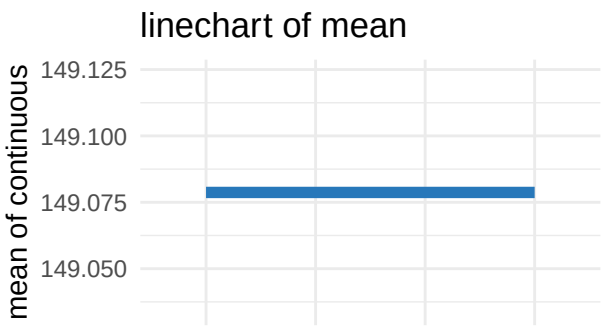


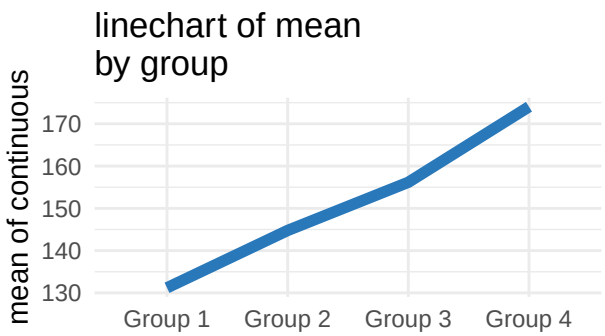

```
my_horiz_m_lollipop_chart <- ggplot(mydata,
                                    aes(x = 1,
                                        y = q,
                                        fill = factor(1))) +
  stat_summary(fun = mean,
               geom = "point",
               size = 5) +
  geom_segment(aes(x = 1,
                   xend = 1,
                   y = 0,
                   yend = mean(q))) +
  coord_flip() +
  scale_color_manual(values = c(arboretum_blue)) +
  ggtitle("horizontal lollipop chart of mean") +
  guides(fill = FALSE) +
  xlab(" ") + ylab("mean of continuous") +
  theme_minimal() +
  theme(axis.text.y = element_blank(),
        axis.ticks.y = element_blank())

my_facet_horiz_m_lollipop_chart <- ggplot(mydata,
                                    aes(x = factor(s),
                                        y = q,
                                        color = s)) +
  stat_summary(fun = mean,
               geom = "point",
               size = 5) +
  geom_segment(aes(x = factor(s),
                   xend = factor(s),
                   y = 0,
                   yend = mean(q))) +
  coord_flip() +
  scale_color_manual(name = "group",
                     values = c(arboretum_blue,
                                taubman_teal,
                                michigan_blue,
                                michigan_maize)) +
  ggtitle("horizontal lollipop chart of mean \nby group") +
  guides(fill=FALSE) +
```

```
  xlab(" ") +
  ylab("mean of continuous") +
  theme_minimal() +
  theme(axis.title.y = element_blank(),
        axis.ticks = element_blank())

plot_grid(my_horiz_m_lollipop_chart,
          my_facet_horiz_m_lollipop_chart)
```

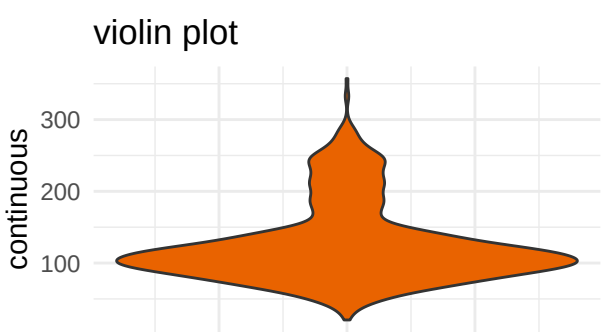


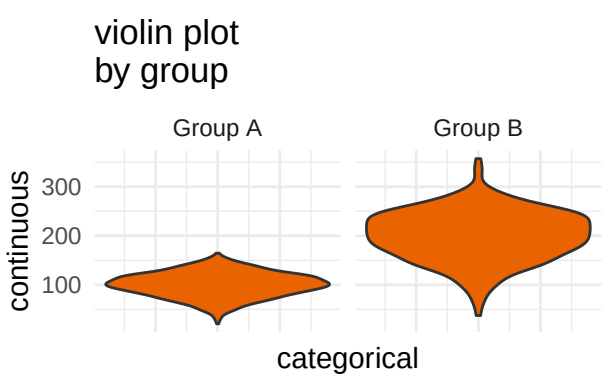

```
my_m_linechart <- ggplot(mydata,
                         aes(x = factor(s),
                             y = mean(q),
                             group = 1)) +
  stat_summary(fun = mean,
               geom = "line",
               size = 2,
               color = arboretum_blue) +
  geom_blank() +
  ggtitle("linechart of mean") +
  xlab(" ") +
  ylab("mean of continuous") +
  theme_minimal() +
  theme(axis.text.x = element_blank()) +
  theme(axis.ticks.x = element_blank())

my_facet_m_linechart <- ggplot(mydata,
                               aes(x = factor(s),
                                   y = q,
                                   group = 1)) +
  stat_summary(fun = mean,
               geom = "line",
               size = 2,
               color = arboretum_blue) +
  ggtitle("linechart of mean \nby group") +
  xlab(" ") +
  ylab("mean of continuous") +
  theme_minimal()

plot_grid(my_m_linechart, my_facet_m_linechart)
```



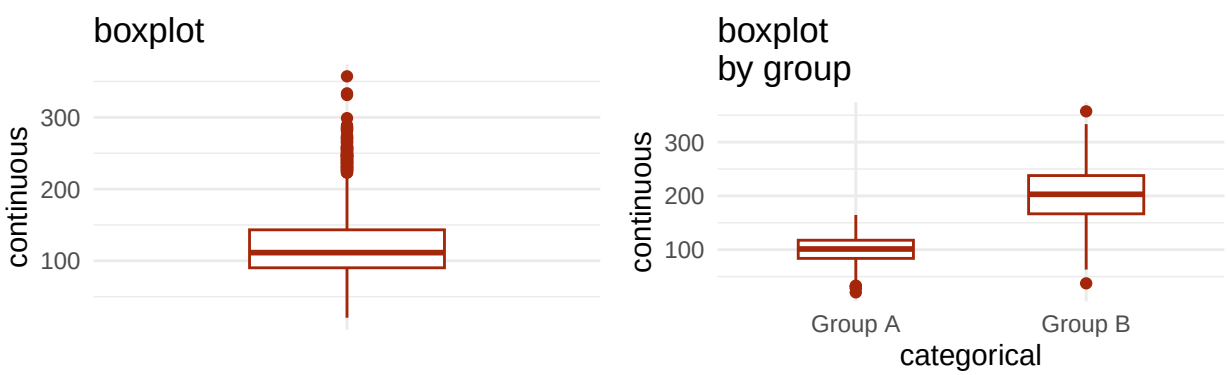



```
my_violin <- ggplot(mydata,
                    aes(x = 1,
                        y = y)) +
  geom_violin(fill = ross_school_orange) +
  ggtitle("violin plot") +
  xlab(" ") +
  ylab("continuous") +
  theme_minimal()  +
  theme(axis.text.x = element_blank()) +
  theme(axis.ticks.x = element_blank())

my_facet_violin <- ggplot(mydata,
                          aes(x = 1,
                              y = y)) +
  geom_violin(fill = ross_school_orange) +
  facet_wrap(~w,
             ncol = 2) +
  ggtitle("violin plot \nby group") +
  xlab("categorical") +
  ylab("continuous") +
  theme_minimal() +
  theme(axis.text.x = element_blank()) +
  theme(axis.ticks.x = element_blank())

plot_grid(my_violin, my_facet_violin, ncol = 2)
```



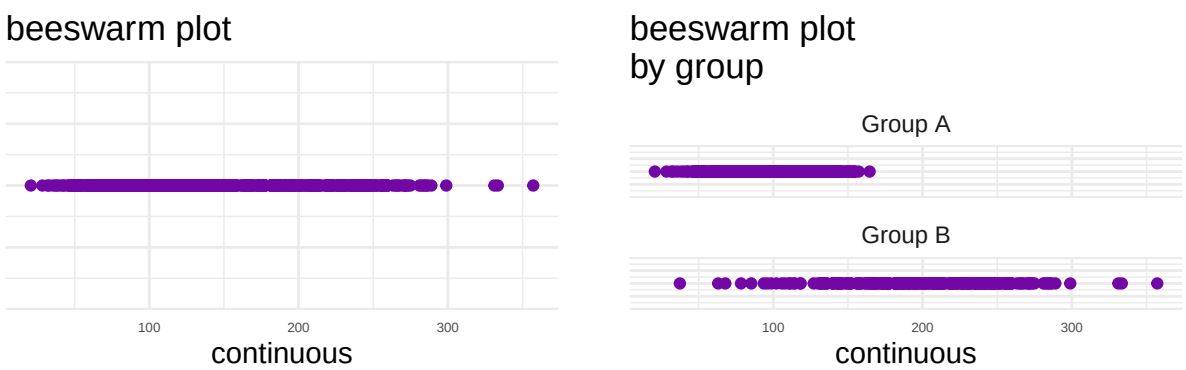


```r
my_boxplot <- ggplot(mydata,
                     aes(x = 2,
                         y = y)) +
  geom_boxplot(colour=tappan_red) +
  scale_x_discrete(limit = c(0,1,2)) +
  ggtitle("boxplot") +
  xlab(" ") +
  ylab("continuous") +
  theme_minimal() +
  theme(axis.text.x = element_blank()) +
  theme(axis.ticks.x = element_blank())

my_conditional_boxplot <- ggplot(mydata,
                                 aes(x = w,
                                     y = y)) +
  geom_boxplot(colour=tappan_red, width = .5) +
  ggtitle("boxplot \nby group") +
  xlab("categorical") +
  ylab("continuous") +
  theme_minimal()

plot_grid(my_boxplot, my_conditional_boxplot, ncol = 2)
```

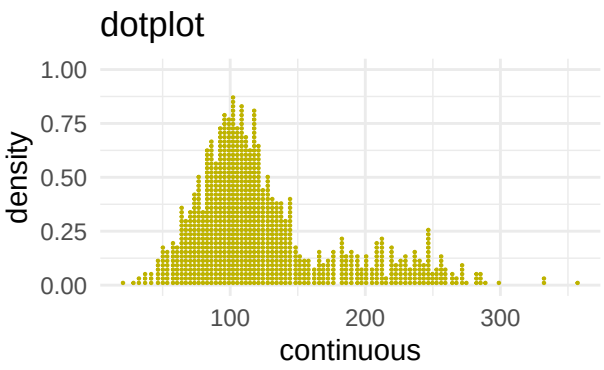


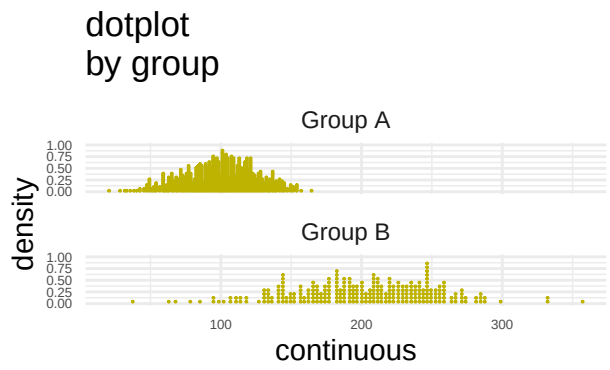

```r
library(ggbeeswarm) # beeswarm geometry

my_beeswarm <- ggplot(mydata,
                      aes(x = y,
                          y = 1)) +
  geom_beeswarm(colour = ann_arbor_amethyst,
                groupOnX = FALSE) +
  ggtitle("beeswarm plot") +
  xlab("continuous") +
  ylab("") +
  theme_minimal() +
  theme(axis.text=element_text(size = 5),
        axis.text.y = element_blank())

my_facet_beeswarm <- ggplot(mydata,
                            aes(x = y,
```

```
                                y = 1)) +
  geom_beeswarm(colour = ann_arbor_amethyst,
                groupOnX = FALSE) +
  facet_wrap(~w, nrow = 2) +
  ggtitle("beeswarm plot \nby group") +
  xlab("continuous") +
  ylab("") +
  theme_minimal() +
  theme(axis.text.x = element_text(size = 5),
        axis.text.y = element_blank())

plot_grid(my_beeswarm, my_facet_beeswarm)
```




```
library(ggdist) # 'distribution' geometries

my_dotplot <- ggplot(mydata,
                     aes(x = y)) +
  stat_dots(colour=wave_field_green) +
  ggtitle("dotplot") +
  xlab("continuous") +
  ylab("density") +
  theme_minimal()

my_facet_dotplot <- ggplot(mydata,
                           aes(x = y)) +
  stat_dots(colour=wave_field_green) +
  facet_wrap(~w, nrow = 2) +
  ggtitle("dotplot \nby group") +
  xlab("continuous") +
  ylab("density") +
  theme_minimal() +
  theme(axis.text = element_text(size = 5)) # small font size for axis

plot_grid(my_dotplot, my_facet_dotplot, ncol = 2)
```





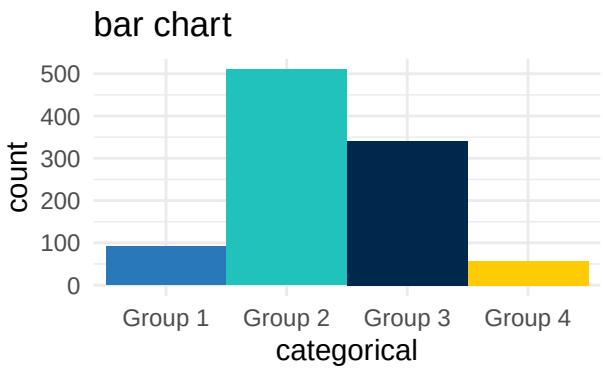
One Thing At A Time

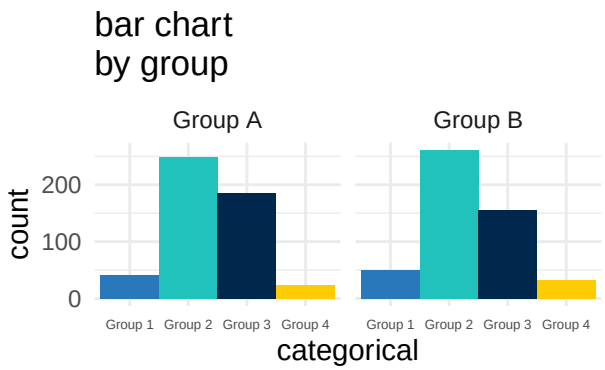
Two Things At A Time

Categorical

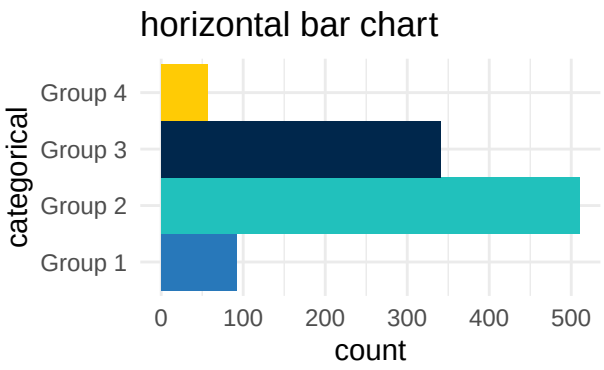
Categorical By Categorical

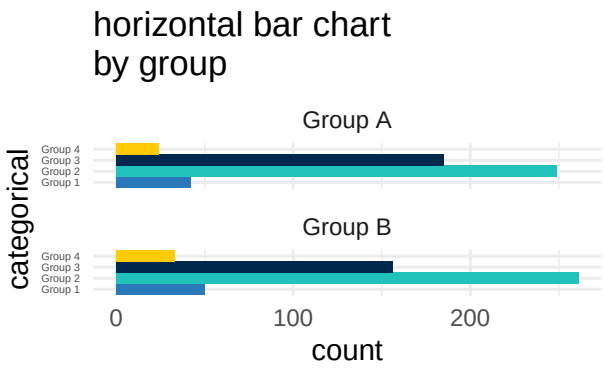
```
my_barchart <- ggplot(mydata,
                      aes(s,
                          fill = s)) +
  geom_bar(width = 1.0) +
  scale_fill_manual(values = c(arboretum_blue,
                               taubman_teal,
                               michigan_blue,
                               michigan_maize)) +
  ggtitle("bar chart") +
  guides(fill=FALSE) +
  xlab("categorical") +
  ylab("count") +
  theme_minimal()

my_facet_barchart <- ggplot(mydata,
                            aes(s,
                                fill = s)) +
  geom_bar(width = 1.0) +
  facet_wrap(~u, ncol = 2) +
  scale_fill_manual(values = c(arboretum_blue,
                               taubman_teal,
                               michigan_blue,
                               michigan_maize)) +
  ggtitle("bar chart \nby group") +
  guides(fill=FALSE) +
  xlab("categorical") + ylab("count") +
  theme_minimal() +
  theme(axis.text.x = element_text(size = rel(.6)))

plot_grid(my_barchart, my_facet_barchart, ncol = 2)
```



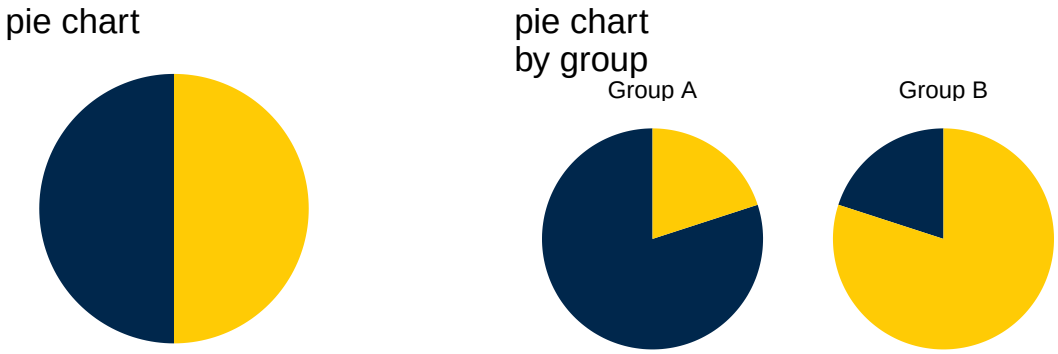



```r
my_horiz_barchat <- ggplot(mydata, aes(s, fill = s)) +
  geom_bar(width = 1.0) +
  coord_flip() +
  scale_fill_manual(values = c(arboretum_blue,
                               taubman_teal,
                               michigan_blue,
                               michigan_maize)) +
  ggtitle("horizontal bar chart") +
  guides(fill=FALSE) +
  xlab("categorical") +
  ylab("count") +
  theme_minimal()

my_horiz_facet_barchart <- ggplot(mydata, aes(s, fill = s)) +
  geom_bar(width = 1.0) +
  facet_wrap(~u, ncol = 1) +
  coord_flip() +
  scale_fill_manual(values = c(arboretum_blue,
                               taubman_teal,
                               michigan_blue,
                               michigan_maize)) +
  ggtitle("horizontal bar chart \nby group") +
  guides(fill=FALSE) +
  xlab("categorical") +
  ylab("count") +
  theme_minimal() +
  theme(axis.text.y = element_text(size = rel(.5)))

plot_grid(my_horiz_barchat, my_horiz_facet_barchart, ncol = 2)
```

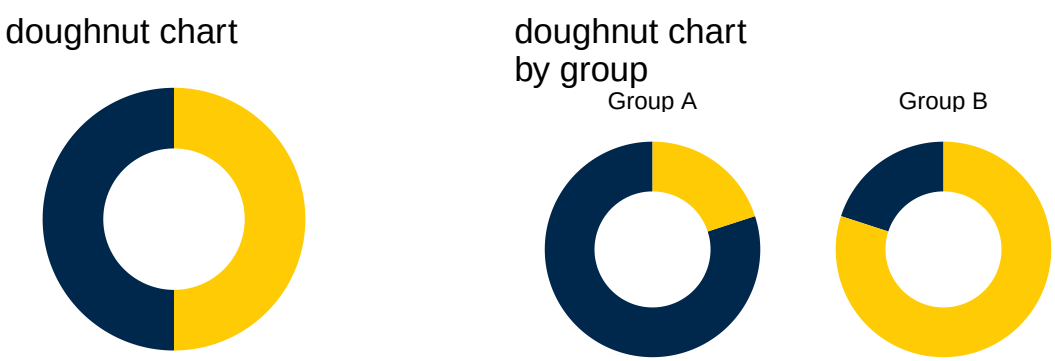





```
my_pie <- ggplot(mydata, aes(x = 1, fill = v)) +
  geom_bar(width = 1) +
  coord_polar(theta="y") +
  scale_fill_manual(values = michigan_colors) +
  ggtitle("pie chart") + guides(fill=FALSE) +
  xlab(" ") +
  ylab("categorical") +
  theme_void() +
  theme(axis.text.x = element_blank()) +
  theme(axis.text.y = element_blank()) +
  theme(axis.ticks = element_blank())

my_facet_pie <- ggplot(mydata, aes(x = 1, fill = v)) +
  geom_bar(width = 1, position = "fill") +
  coord_polar(theta="y") +
  facet_wrap(~u, ncol = 2) +
  scale_fill_manual(values = michigan_colors) +
  ggtitle("pie chart \nby group") +
  guides(fill=FALSE) +
  xlab(" ") +
  ylab("categorical") +
  theme_void() +
  theme(axis.text.x = element_blank()) +
  theme(axis.text.y = element_blank())  +
  theme(axis.ticks = element_blank())

plot_grid(my_pie, my_facet_pie, ncol = 2)
```



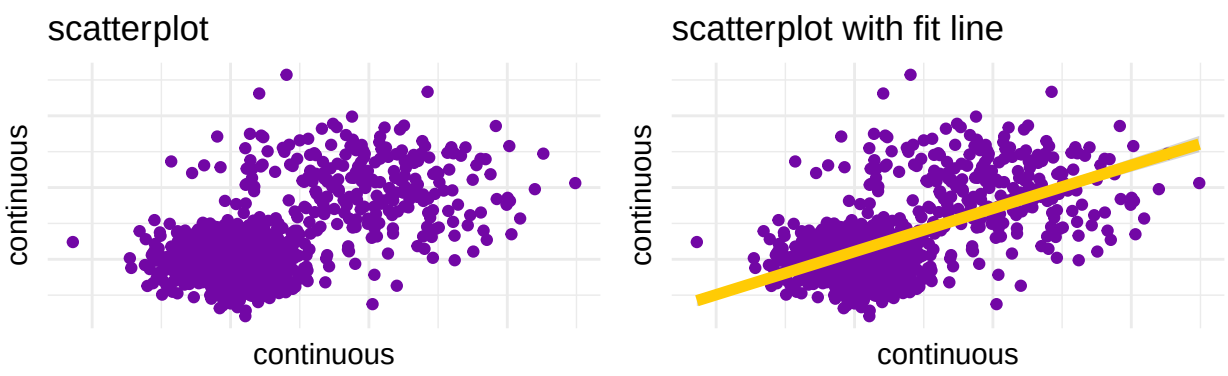
```r
my_doughnut <- ggplot(mydata, aes(x = 2, fill = v)) +
  geom_bar() +
  coord_polar(theta="y") +
  scale_fill_manual(values = michigan_colors) +
  ggtitle("doughnut chart") +
  guides(fill=FALSE) +
  xlab(" ") +
  ylab("categorical") +
  xlim(.5, 2.5) +
  theme_void() +
  theme(axis.text.x = element_blank()) +
  theme(axis.text.y = element_blank()) +
  theme(axis.ticks = element_blank())

my_facet_doughnut <- ggplot(mydata, aes(x = 2, fill = v)) +
  geom_bar(position = "fill") +
  coord_polar(theta="y") +
  facet_wrap(~u, ncol = 2) +
  scale_fill_manual(values = michigan_colors) +
  ggtitle("doughnut chart \nby group") +
  guides(fill=FALSE) +
  xlab(" ") +
  ylab("categorical") +
  xlim(.5, 2.5) +
  theme_void() +
  theme(axis.text.x = element_blank()) +
  theme(axis.text.y = element_blank()) +
  theme(axis.ticks = element_blank())

plot_grid(my_doughnut, my_facet_doughnut, ncol = 2)
```



Continuous by Continuous

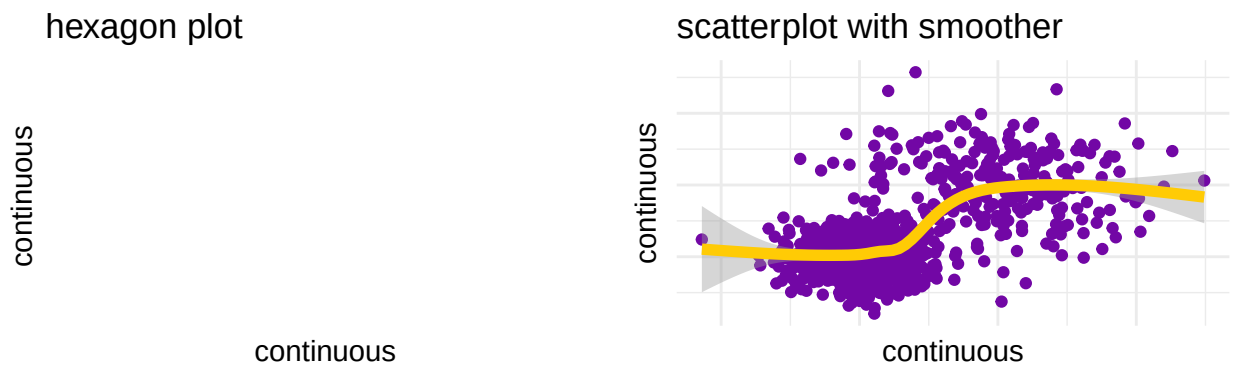
```r
my_scatterplot <- ggplot(mydata, aes(x=x, y=y)) +
  geom_point(colour = ann_arbor_amethyst) +
  ggtitle("scatterplot") +
  xlab("continuous") +
```

```
  ylab("continuous") +
  theme_minimal() +
  theme(axis.text.x = element_blank()) +
  theme(axis.text.y = element_blank()) +
  theme(axis.ticks = element_blank())

my_scatterplot_smoother <- ggplot(mydata, aes(x=x, y=y)) +
  geom_point(colour = ann_arbor_amethyst) +
  geom_smooth(method = lm, color = michigan_maize, size = 2) +
  ggtitle("scatterplot with fit line") +
  xlab("continuous") +
  ylab("continuous") +
  theme_minimal() +
  theme(axis.text.x = element_blank()) +
  theme(axis.text.y = element_blank()) +
  theme(axis.ticks = element_blank())

plot_grid(my_scatterplot, my_scatterplot_smoother)
```

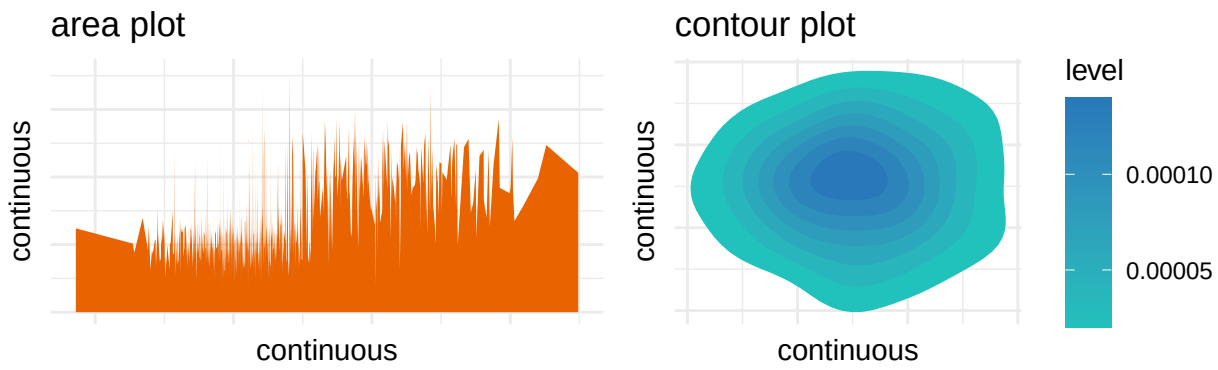



```
my_hexagon <- ggplot(mydata, aes(x=x, y=y)) +
  geom_hex() +
  ggtitle("hexagon plot") +
  xlab("continuous") +
  ylab("continuous") +
  theme_minimal() +
  theme(legend.text = element_text(size=4),
        legend.key.size = unit(.25, "cm")) +
  theme(axis.text.x = element_blank()) +
  theme(axis.text.y = element_blank()) +
  theme(axis.ticks = element_blank()) +
  scale_fill_gradient(low = taubman_teal,
                      high = arboretum_blue)

my_smoother <- ggplot(mydata, aes(x=x, y=y)) +
  geom_point(colour = ann_arbor_amethyst) +
  geom_smooth(se=TRUE, color=michigan_maize, size=2) +
  ggtitle("scatterplot with smoother") +
  xlab("continuous") +
  ylab("continuous") +
```

```
  theme_minimal() +
  theme(axis.text.x = element_blank()) +
  theme(axis.text.y = element_blank())  +
  theme(axis.ticks = element_blank())

plot_grid(my_hexagon, my_smoother)
```




```
my_area <- ggplot(mydata, aes(x=x, y=y)) +
  geom_area(position = "stack", fill = ross_school_orange) +
  ggtitle("area plot") +
  xlab("continuous") +
  ylab("continuous") +
  theme_minimal() +
  theme(axis.text.x = element_blank()) +
  theme(axis.text.y = element_blank())  +
  theme(axis.ticks = element_blank())

my_contour <- ggplot(mydata, aes(x=x, y=y)) +
  stat_density_2d(aes(fill = ..level..),
                  geom = "polygon") +
  ggtitle("contour plot") +
  xlab("continuous") +
  ylab("continuous") +
  theme_minimal() +
  theme(axis.text.x = element_blank()) +
  theme(axis.text.y = element_blank())  +
  theme(axis.ticks = element_blank()) +
  scale_fill_gradient(low = taubman_teal,
                      high = arboretum_blue)

plot_grid(my_area, my_contour)
```



Graphics made with the ggplot2 graphing library created by Hadley Wickham.

Available online at https://agrogan1.github.io/

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